Equation Reference for stellar2d

All numbered equations in this document are referenced by source code comments of the form `// Eq. (X.Y)`. This is the authoritative specification — if code disagrees with this document, the code has a bug.

Coordinate system: axisymmetric spherical (r, θ), with $\partial/\partial\phi = 0$ and $v_\phi = 0$. $\theta \in [0, \pi]$ is the polar angle measured from the north pole.

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§1 Governing Equations

Conservative variables

$$\mathbf{U} = \begin{pmatrix} \rho \\ \rho v_r \\ \rho v_\theta \\ \rho E \end{pmatrix} \tag{1.1}$$

where total specific energy $E = e + \frac{1}{2}(v_r^2 + v_\theta^2)$ and e is specific internal energy.

Ideal gas equation of state

$$P = (\gamma - 1)\,\rho\, e \tag{1.2}$$

Sound speed

$$c_s = \sqrt{\gamma\, P/\rho} \tag{1.3}$$

Euler equations in conservative form (spherical, axisymmetric)

$$\frac{\partial \mathbf{U}}{\partial t} + \frac{1}{r^2}\frac{\partial (r^2 \mathbf{F})}{\partial r} + \frac{1}{r\sin\theta}\frac{\partial(\sin\theta\, \mathbf{G})}{\partial \theta} = \mathbf{S} \tag{1.4}$$

Radial flux:

$$\mathbf{F} = \begin{pmatrix} \rho v_r \\ \rho v_r^2 + P \\ \rho v_r v_\theta \\ (\rho E + P)\, v_r \end{pmatrix} \tag{1.5}$$

Theta flux:

$$\mathbf{G} = \begin{pmatrix} \rho v_\theta \\ \rho v_r v_\theta \\ \rho v_\theta^2 + P \\ (\rho E + P)\, v_\theta \end{pmatrix} \tag{1.6}$$

Geometric + gravity source:

$$\mathbf{S} = \begin{pmatrix} 0 \\ \dfrac{\rho v_\theta^2}{r} + \dfrac{2P}{r} - \rho \dfrac{\partial \Phi}{\partial r} \\ \dfrac{P \cot\theta}{r} - \dfrac{\rho v_r v_\theta}{r} - \dfrac{\rho}{r}\dfrac{\partial \Phi}{\partial \theta} \\ -\rho\, \mathbf{v} \cdot \nabla\Phi \end{pmatrix} \tag{1.7}$$

Note: the $2P/r$ and $P\cot\theta/r$ terms are geometric consequences of expressing the divergence in spherical coordinates, not additional physics.

Poisson equation (self-gravity)

$$\nabla^2 \Phi = 4\pi G \rho \tag{1.8}$$

Expanded in axisymmetric spherical coordinates:

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2 \frac{\partial \Phi}{\partial r}\right) + \frac{1}{r^2 \sin\theta}\frac{\partial}{\partial \theta}\left(\sin\theta\, \frac{\partial \Phi}{\partial \theta}\right) = 4\pi G \rho \tag{1.9}$$

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§2 Finite Volume Discretization

Structured grid with $N_r \times N_\theta$ cells. Indices: $i = 0, \dots, N_r - 1$ (radial), $j = 0, \dots, N_\theta - 1$ (polar). Cell interfaces at $r_{i+1/2}$, $\theta_{j+1/2}$.

Logarithmic radial mesh

$$r_{i+1/2} = R_{\text{outer}} \left(\frac{i}{N_r}\right)^{\alpha} \tag{2.1}$$

Cell volume (azimuthal 2π factor omitted; cancels in all FV updates)

$$V_{ij} = \frac{r_{i+1/2}^3 - r_{i-1/2}^3}{3}\left(\cos\theta_{j-1/2} - \cos\theta_{j+1/2}\right) \tag{2.2}$$

Radial face area

$$A^r_{i+1/2,\,j} = r_{i+1/2}^2 \left(\cos\theta_{j-1/2} - \cos\theta_{j+1/2}\right) \tag{2.3}$$

Theta face area

$$A^\theta_{i,\,j+1/2} = \sin\theta_{j+1/2}\; \frac{r_{i+1/2}^2 - r_{i-1/2}^2}{2} \tag{2.4}$$

Semi-discrete finite volume update

$$\frac{d\mathbf{U}_{ij}}{dt} = -\frac{1}{V_{ij}}\Big[A^r_{i+1/2,j}\,\hat{\mathbf{F}}_{i+1/2,j} - A^r_{i-1/2,j}\,\hat{\mathbf{F}}_{i-1/2,j} + A^\theta_{i,j+1/2}\,\hat{\mathbf{G}}_{i,j+1/2} - A^\theta_{i,j-1/2}\,\hat{\mathbf{G}}_{i,j-1/2}\Big] + \mathbf{S}_{ij} \quad (2.5)$$

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§3 MUSCL Reconstruction

Minmod limiter

$$\text{minmod}(a,b) = \begin{cases} \text{sgn}(a)\min(|a|,|b|) & \text{if } ab > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.1)$$

Van Leer limiter

$$\text{vanleer}(a,b) = \begin{cases} \dfrac{2ab}{a+b} & \text{if } ab > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.2)$$

Limited slope at cell i

$$\sigma_i = \text{limiter}(w_i - w_{i-1},\ w_{i+1} - w_i) \quad (3.3)$$

Reconstructed left state at face $i+1/2$

$$w^L_{i+1/2} = w_i + \tfrac{1}{2}\,\sigma_i \quad (3.4)$$

Reconstructed right state at face $i+1/2$

$$w^R_{i+1/2} = w_{i+1} - \tfrac{1}{2}\,\sigma_{i+1} \quad (3.5)$$

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§4 HLLC Riemann Solver

Reference: Toro, "Riemann Solvers and Numerical Methods for Fluid Dynamics", 3rd ed., §10.4.

Wave speed estimates

$$S_L = \min(u_L - c_L,\ u_R - c_R), \qquad S_R = \max(u_L + c_L,\ u_R + c_R) \quad (4.1)$$

Contact wave speed

$$S^* = \frac{P_R - P_L + \rho_L u_L(S_L - u_L) - \rho_R u_R(S_R - u_R)}{\rho_L(S_L - u_L) - \rho_R(S_R - u_R)} \tag{4.2}$$

Total energy density

$$\mathcal{E} = \frac{P}{\gamma - 1} + \frac{1}{2}\rho(u^2 + v_t^2) \tag{4.3}$$

Physical flux (in the normal direction u, tangential v_t)

$$\mathbf{F} = \begin{pmatrix} \rho u \\ \rho u^2 + P \\ \rho u\, v_t \\ (\mathcal{E} + P)\, u \end{pmatrix} \tag{4.4}$$

For radial faces: $u = v_r$, $v_t = v_\theta$. For theta faces: $u = v_\theta$, $v_t = v_r$.

HLLC star-state density

$$\rho_K^* = \rho_K \frac{S_K - u_K}{S_K - S^*} \tag{4.5}$$

HLLC star-state energy

$$\mathcal{E}_K^* = \rho_K^* \left[\frac{\mathcal{E}_K}{\rho_K} + (S^* - u_K)\left(S^* + \frac{P_K}{\rho_K(S_K - u_K)} \right) \right] \tag{4.6}$$

HLLC numerical flux

$$\hat{\mathbf{F}} = \begin{cases} \mathbf{F}_L & \text{if } S_L \geq 0 \\ \mathbf{F}_L + S_L(\mathbf{U}_L^* - \mathbf{U}_L) & \text{if } S_L < 0 \leq S^* \\ \mathbf{F}_R + S_R(\mathbf{U}_R^* - \mathbf{U}_R) & \text{if } S^* < 0 \leq S_R \\ \mathbf{F}_R & \text{if } S_R < 0 \end{cases} \tag{4.7}$$

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§5 Geometric Source Terms

These arise from expanding $\nabla\cdot$ in spherical coordinates. They must be discretized in a **volume-consistent** manner to achieve well-balanced equilibrium (see §5.3).

Radial momentum geometric source (continuous)

$$S_{m_r}^{\text{geom}} = \frac{\rho v_\theta^2}{r} + \frac{2P}{r} \tag{5.1}$$

Theta momentum geometric source (continuous)

$$S_{m_\theta}^{\text{geom}} = \frac{P\cot\theta}{r} - \frac{\rho v_r v_\theta}{r} \tag{5.2}$$

Volume-consistent discrete geometric source (well-balanced)

The continuous $2P/r$ source must be discretized consistently with the FV divergence operator. We replace the point-value $2P/r$ with its volume average:

$$\left\langle \frac{2P}{r} \right\rangle_{ij} = \frac{1}{V_{ij}} \int_{\text{cell}} \frac{2P}{r}\, r^2 \sin\theta\, dr\, d\theta = \frac{P_{ij}}{V_{ij}} \left(r_{i+1/2}^2 - r_{i-1/2}^2\right)\left(\cos\theta_{j-1/2} - \cos\theta_{j+1/2}\right) \tag{5.3}$$

(assuming P approximately uniform within the cell).

Similarly, the $P\cot\theta/r$ source:

$$\left\langle \frac{P\cot\theta}{r} \right\rangle_{ij} = \frac{P_{ij}}{V_{ij}} \frac{r_{i+1/2}^2 - r_{i-1/2}^2}{2} \left(\sin\theta_{j-1/2} - \sin\theta_{j+1/2}\right) \tag{5.4}$$

(derived from $\int \cos\theta\, d\theta = \sin\theta$).

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§6 Gravity

Gravity source — radial momentum

$$S_{m_r}^{g} = -\rho \frac{\partial \Phi}{\partial r} \tag{6.1}$$

Gravity source — theta momentum

$$S_{m_\theta}^{g} = -\frac{\rho}{r} \frac{\partial \Phi}{\partial \theta} \tag{6.2}$$

Gravity source — energy

$$S_E^{g} = -\rho \left(v_r \frac{\partial \Phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial \Phi}{\partial \theta} \right) \tag{6.3}$$

Discrete Laplacian — radial part

$$L_{ij}^r = \frac{1}{r_i^2} \frac{r_{i+1/2}^2 \dfrac{\Phi_{i+1,j} - \Phi_{ij}}{\delta r_i^+} - r_{i-1/2}^2 \dfrac{\Phi_{ij} - \Phi_{i-1,j}}{\delta r_i^-}}{\Delta r_i} \tag{6.4}$$

where $\delta r_i^+ = r_{i+1}^c - r_i^c$, $\delta r_i^- = r_i^c - r_{i-1}^c$, $\Delta r_i = r_{i+1}^f - r_i^f$ (face-to-face width).

Discrete Laplacian — theta part

$$L_{ij}^\theta = \frac{1}{r_i^2 \sin\theta_j} \frac{\sin\theta_{j+1/2} \dfrac{\Phi_{i,j+1} - \Phi_{ij}}{\delta\theta_j^+} - \sin\theta_{j-1/2} \dfrac{\Phi_{ij} - \Phi_{i,j-1}}{\delta\theta_j^-}}{\Delta\theta_j} \tag{6.5}$$

where $\delta\theta_j^\pm$ are center-to-center distances, $\Delta\theta_j$ is face-to-face width.

Poisson boundary conditions

Dirichlet at outer boundary (monopole approximation):

$$\Phi(R_{\text{outer}}) = -\frac{G\, M_{\text{total}}}{R_{\text{outer}}} \tag{6.6}$$

Neumann at center ($r = 0$, symmetry):

$$\left.\frac{\partial\Phi}{\partial r}\right|_{r=0} = 0 \tag{6.7}$$

Neumann at poles ($\theta = 0, \pi$, axial symmetry):

$$\left.\frac{\partial\Phi}{\partial\theta}\right|_{\theta=0,\pi} = 0 \tag{6.8}$$

Gravity gradient at cell center (face-based interpolation)

Face gradients:

$$\left.\frac{\partial\Phi}{\partial r}\right|_{i+1/2} = \frac{\Phi_{i+1,j} - \Phi_{ij}}{r_{i+1}^c - r_i^c} \tag{6.9a}$$

Cell-center gradient by weighted average of flanking face gradients:

$$\left.\frac{\partial\Phi}{\partial r}\right|_i = \frac{\delta r_i^+ \cdot g_{i-1/2} + \delta r_i^- \cdot g_{i+1/2}}{\delta r_i^+ + \delta r_i^-} \tag{6.9b}$$

At $i = 0$: use $g_{i-1/2} = 0$ (Neumann, Eq. 6.7). At $i = N_r - 1$: use the Dirichlet value (Eq. 6.6) to form $g_{i+1/2}$.
Theta gradient analogously:

$$\left.\frac{\partial \Phi}{\partial \theta}\right|_{j+1/2} = \frac{\Phi_{i,j+1} - \Phi_{ij}}{\theta^c_{j+1} - \theta^c_j} \tag{6.10a}$$

$$\left.\frac{\partial \Phi}{\partial \theta}\right|_{j} = \frac{\delta\theta^+_j \cdot g^\theta_{j-1/2} + \delta\theta^-_j \cdot g^\theta_{j+1/2}}{\delta\theta^+_j + \delta\theta^-_j} \tag{6.10b}$$

At $j = 0$ and $j = N_\theta - 1$: use $g^\theta = 0$ (Neumann, Eq. 6.8).

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§7 Time Integration

RK2 — Heun's method

Stage 1:

$$\mathbf{U}^* = \mathbf{U}^n + \Delta t\, \mathbf{R}(\mathbf{U}^n) \tag{7.1}$$

Stage 2:

$$\mathbf{U}^{**} = \mathbf{U}^* + \Delta t\, \mathbf{R}(\mathbf{U}^*) \tag{7.2}$$

Average:

$$\mathbf{U}^{n+1} = \tfrac{1}{2}(\mathbf{U}^n + \mathbf{U}^{**}) \tag{7.3}$$

where $\mathbf{R}(\mathbf{U})$ includes flux divergence, geometric source, and gravity source.

CFL condition

$$\Delta t = C_{\text{CFL}} \cdot \min_{i,j} \left[\min\left(\frac{\Delta r_i}{|v_r| + c_s},\ \frac{r_i\, \Delta\theta_j}{|v_\theta| + c_s} \right) \right] \tag{7.4}$$

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§8 Boundary Conditions

Inner boundary ($r = 0$): reflecting symmetry

For ghost cell at layer g ($g = 1, 2, \ldots$):

$$\rho_{-g,j} = \rho_{g-1,j}, \quad (\rho v_r)_{-g,j} = -(\rho v_r)_{g-1,j}, \quad (\rho v_\theta)_{-g,j} = (\rho v_\theta)_{g-1,j}, \quad (\rho E)_{-g,j} = (\rho E)_{g-1,j} \tag{8.1}$$

Outer boundary ($r = R$): zero-gradient outflow

$$\mathbf{U}_{N_r+g,\,j} = \mathbf{U}_{N_r-1,\,j} \tag{8.2}$$

Axis boundaries ($\theta = 0$ and $\theta = \pi$): reflecting symmetry in v_θ

For $\theta = 0$ (north pole), ghost layer g:

$$\rho_{i,-g} = \rho_{i,g-1}, \quad (\rho v_r)_{i,-g} = (\rho v_r)_{i,g-1}, \quad (\rho v_\theta)_{i,-g} = -(\rho v_\theta)_{i,g-1}, \quad (\rho E)_{i,-g} = (\rho E)_{i,g-1} \tag{8.3}$$

South pole ($\theta = \pi$) identical with indices mirrored from $j = N_\theta - 1$.

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§9 Initial Conditions

Lane-Emden equation

$$\frac{1}{\xi^2}\frac{d}{d\xi}\left(\xi^2\frac{d\Theta}{d\xi}\right) + \Theta^n = 0, \qquad \Theta(0) = 1, \quad \Theta'(0) = 0 \tag{9.1}$$

Lane-Emden length scale

$$\alpha^2 = \frac{(n+1)\,K\,\rho_c^{1/n-1}}{4\pi G} \tag{9.2}$$

Stellar radius

$$R_\star = \alpha\,\xi_1 \tag{9.3}$$

where ξ_1 is the first zero of $\Theta(\xi)$.

Density from Lane-Emden solution

$$\rho(r) = \rho_c\,\Theta\Big(\frac{r}{\alpha}\Big)^n \tag{9.4}$$

Polytropic pressure

$$P = K\,\rho^{1+1/n} \tag{9.5}$$

Sedov blast energy deposition

$$P_{\text{blast}} = (\gamma - 1)\,\rho_0\,\frac{E_{\text{blast}}}{V_{\text{blast}}} \tag{9.6}$$

where V_{blast} uses the same volume convention as the code (no 2π azimuthal factor).

Evrard collapse density profile

$$\rho(r) = \frac{M}{2\pi R^3}\,\frac{1}{r}, \qquad r < R \tag{9.7}$$

Jeans instability perturbation

$$\rho = \rho_0\big[1 + \epsilon\cos(k_r r)\cos(k_\theta \theta)\big] \tag{9.8}$$

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§10 Low-Mach Implicit Solver (JFNK)

The low-Mach solver uses Backward Euler with Jacobian-Free Newton-Krylov (JFNK) to take time steps unconstrained by the acoustic CFL.

Backward Euler nonlinear system

$$F(U^{n+1}) = \frac{U^{n+1} - U^n}{\Delta t} - R(U^{n+1}) = 0 \tag{10.1}$$

where $R(U)$ is the spatial residual (§1, §5, §6) using 1st-order upwind advection and central-difference pressure/gravity gradients.

Newton iteration

$$J_k\,\delta U = -F(U^k), \qquad U^{k+1} = U^k + \alpha\,\delta U \tag{10.2}$$

with backtracking line search on the scaled merit function $\|L^{-1}F\|_2$.

Jacobian-free matrix-vector product

$$J\,v \approx \frac{F(U^k + \varepsilon v) - F(U^k)}{\varepsilon}, \qquad \varepsilon = \sqrt{\epsilon_{\text{mach}}}\,\frac{1 + \|U\|}{\|v\|} \tag{10.3}$$

FGMRES linear solver

Right-preconditioned FGMRES(120) with Eisenstat-Walker adaptive forcing (initial $\eta = 10^{-2}$, EW Choice 2).

1D radial gravity

Gravity is computed from angle-averaged density via cumulative mass integral (no Poisson solve, no GMG noise):

$$g_r(r_i) = -\frac{G\,M(<r_i)}{r_i^2}, \qquad M(<r_i) = \sum_{k<i} \bar{\rho}_k\, V_k \tag{10.4}$$

where $\bar{\rho}_k$ is the θ-averaged density at shell k.

Well-balanced residual (reference-state subtraction)

The momentum residual uses perturbation form to achieve $R(U_{\text{HSE}}) = 0$ to machine precision:

$$F_r = -\nabla P' + \rho'\, g_0(r) + \rho\, g'(r) + S_{\text{geom}} \tag{10.5}$$

where primed quantities are deviations from the HSE reference state.

Atmosphere treatment

Cells with $\rho_0 < 10^{-6}\,\rho_{\text{max}}$ are treated as atmosphere: $R = 0$ in the residual kernel, so Newton does not evolve them.

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§11 Physics-Based Preconditioner (PBP)

The JFNK preconditioner $M \approx J$ uses the block structure of the Jacobian to incorporate a pressure Poisson solve, enabling GMRES convergence in $O(10)$ iterations regardless of Δt.

Jacobian block structure

The 4-DOF Jacobian has the block form:

$$J = \begin{pmatrix} A_\rho & 0 & 0 & 0 \\ g_0 & A_{vr} & 0 & B_r \\ 0 & 0 & A_{v\theta} & B_\theta \\ 0 & C_r & C_\theta & A_E \end{pmatrix} \tag{11.1}$$

where $A_v = -(1/\Delta t + \text{advection rate})$ (diagonal-dominant), $B = -(\gamma - 1)\nabla$ (pressure gradient), and $C = -P\,\nabla \cdot (1/\rho)$ (compression work).

Schur complement

Eliminating velocity from the energy equation gives the pressure Schur complement:

$$S = A_E - C\,A_v^{-1}\,B \approx \nabla \cdot \left(\frac{\gamma - 1}{A_p} \nabla \right) \tag{11.2}$$

which is a variable-coefficient Poisson operator, solved by geometric multigrid (GMG).

PBP preconditioner application ($y = M^{-1}x$)

Given input $x = (x_\rho, x_{mr}, x_{m\theta}, x_E)$:
Step 1 — Momentum predict (2-DOF block-tridiagonal r-line solve):

$$\tilde{v}_r, \tilde{v}_\theta = A_v^{-1}\,(x_{mr}, x_{m\theta})/\rho \tag{11.3}$$

Step 2 — Pressure Poisson (GMG, 3 V-cycles):

$$\nabla \cdot \left(\frac{1}{A_p} \nabla\, \delta p \right) = \nabla \cdot \tilde{v} \tag{11.4}$$

Step 3 — Velocity correct and assemble:

$$y_{mr} = \rho \left(\tilde{v}_r - \frac{1}{A_p} \frac{\partial\, \delta p}{\partial r} \right), \qquad y_{m\theta} = \rho \left(\tilde{v}_\theta - \frac{1}{A_p} \frac{1}{r} \frac{\partial\, \delta p}{\partial \theta} \right) \tag{11.5}$$

$$y_\rho = (J^{-1})_{00}\, x_\rho, \qquad y_E = (J^{-1})_{33}\, x_E \tag{11.6}$$

where $(J^{-1})_{00}$ and $(J^{-1})_{33}$ are diagonal elements of the block-Jacobi inverse (point Jacobi on ρ and energy).

Why this works

The preconditioner runs one cycle of the projection method *inside* the Krylov solver. The GMG pressure Poisson directly resolves the global pressure coupling that GMRES alone would need $O(N)$ iterations to discover. Because this is only a preconditioner, approximation errors do not affect the final solution — the outer JFNK iteration converges to the exact Backward Euler solution regardless of preconditioner quality.

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§12 FAS Nonlinear Multigrid (Polar Grid)

The FAS (Full Approximation Scheme) solver operates on the same polar grid as the explicit solver (§1–§8), using nonlinear multigrid with block-Jacobi smoothing.

Well-balanced flux-level P□ subtraction

The momentum flux divergence subtracts background pressure at each face to achieve $R(U_{\text{HSE}}) = 0$ to machine precision. For regular cells ($i \geq 1$):

$$(\nabla{\cdot}F)_{m_r} = -\frac{1}{V_{ij}}\Big[A^r_{i+1/2}(\widehat{F}^{m_r}_{i+1/2}-P^0_{i+1/2})-A^r_{i-1/2}(\widehat{F}^{m_r}_{i-1/2}-P^0_{i-1/2})+A^\theta_{j+1/2}\widehat{G}^{m_r}_{j+1/2}-A^\theta_{j-1/2}\widehat{G}^{m_r}_{j-1/2}\Big] \tag{12.1}$$

where $P^0_{i+1/2} = \frac{1}{2}(P_0(r_i) + P_0(r_{i+1}))$ is the face-averaged HSE pressure. The theta momentum equation subtracts P^0 from the θ-face flux analogously.

Well-balanced gravity source (perturbation split)

$$S^g_{m_r} = \rho'\, g_{0,r} + \rho\, g'_r, \qquad \rho' = \rho - \rho_0, \quad g' = g - g_0 \tag{12.2}$$

where ρ_0, g_0 are the HSE reference state. At equilibrium, $\rho' = 0$ and $g' = 0$, so the source vanishes exactly.

HSE defect subtraction

On coarse grids the discrete flux-level WB is not exact due to averaging. The solver precomputes and subtracts the frozen defect:

$$R_{\text{corrected}}(U) = R(U) - R(U_0) \tag{12.3}$$

ensuring $R_{\text{corrected}}(U_0) = 0$ on every grid level.

Origin cell ($i = 0$)

The innermost cell is a pie-slice wedge touching $r = 0$. It has no inner radial face (zero area). Geometry:

$$V_{0j} = \tfrac{1}{3}\, r^3_{1/2}\, (\cos\theta_{j-1/2} - \cos\theta_{j+1/2}), \qquad A^r_{1/2,j} = r^2_{1/2}\, (\cos\theta_{j-1/2} - \cos\theta_{j+1/2}) \tag{12.4}$$

Volume-averaged $\langle 1/r\rangle$ for geometric source:

$$\left\langle \frac{1}{r} \right\rangle_{0j} = \frac{3}{2\, r_{1/2}} \tag{12.5}$$

First-order HLLC is used at the origin (stencil too short for MUSCL). WB P□ subtraction applies to the single outer radial face:

$$(\nabla \cdot F)_{m_r}\big|_{i=0} = -\frac{1}{V_{0j}}\Big[A^r_{1/2}(\widehat{F}^{m_r}_{1/2} - P^0_{1/2}) + A^\theta_{j+1/2}\widehat{G}^{m_r}_{j+1/2} - A^\theta_{j-1/2}\widehat{G}^{m_r}_{j-1/2}\Big] \tag{12.6}$$

MUSCL reconstruction (perturbation form)

Slopes are computed on perturbation variables, then face states are restored using the face-centered HSE background:

$$\sigma_i^\rho = \text{limiter}(\rho_i' - \rho_{i-1}',\ \rho_{i+1}' - \rho_i') \tag{12.7a}$$

$$\rho_{i+1/2}^L = \rho_0(r_{i+1/2}) + \rho_i' + \tfrac{1}{2}\sigma_i^\rho, \qquad \rho_{i+1/2}^R = \rho_0(r_{i+1/2}) + \rho_{i+1}' - \tfrac{1}{2}\sigma_{i+1}^\rho \tag{12.7b}$$

where $\rho_0(r_{i+1/2}) = \frac{1}{2}(\rho_{0,i} + \rho_{0,i+1})$ is the face-centered background. Pressure is treated identically.

FAS restrict (volume-weighted)

Fine-to-coarse state transfer with 2×2 coarsening:

$$U_{I,J}^H = \frac{\sum_{(i,j)\in\mathcal{C}(I,J)} U_{ij}^h V_{ij}^h}{\sum_{(i,j)\in\mathcal{C}(I,J)} V_{ij}^h} \tag{12.8}$$

where $\mathcal{C}(I,J)$ denotes the 2×2 fine children of coarse cell (I,J).

FAS tau correction

The coarse-level FAS right-hand side incorporates the restricted fine-level defect so that the coarse solve drives the fine-level residual to zero:

$$\tau_H = \frac{U_H}{\Delta t} - R(U_H) + \bar{d}_H \tag{12.9}$$

where $\bar{d}_H$ is the restricted fine defect (Eq. 12.8 applied to $F(U^h) = R(U^h) - U^h/\Delta t + \tau^h$).

FAS prolongate (limited injection)

Coarse correction is piecewise-constant, clamped for safety:

$$\delta U_H = U_H^{\text{solved}} - U_H^{\text{restricted}} \tag{12.10a}$$

$$U_{ij}^h \mathrel{+}= \text{clamp}(\delta U_H,\ -\beta\, s_{ij},\ +\beta\, s_{ij}), \qquad \beta = 0.5 \tag{12.10b}$$

where s_{ij} is a local scale (ρ, ρc_s, or ρE depending on the equation).

Block-Jacobi smoother

$$U^{k+1} = U^k - \omega\, J_{\text{diag}}^{-1}\, F(U^k), \qquad \omega = 0.8 \tag{12.11}$$

The 4×4 diagonal Jacobian block at each cell is:

$$J_{\text{diag}} = -\Big(\frac{1}{\Delta t} + s_r\Big) I + \text{off-diagonal couplings} \tag{12.12}$$

where $s_r = 2\big[(|v_r| + c_s)/\Delta r + (|v_\theta| + c_s)/(r\Delta\theta)\big]$ is the spectral radius estimate (factor 2 for MUSCL doubling). Off-diagonal entries capture pressure–energy coupling ($\partial(\nabla P)/\partial(\rho E)$) and gravity–density coupling ($\partial S^g/\partial\rho$). The block is inverted per cell via 4×4 Gauss–Jordan elimination.

Smooth floor

Density and internal energy are kept non-negative by a C^∞ smooth approximation to max(x, ε):

$$\text{floor}(x) = \tfrac{1}{2}(x + \sqrt{x^2 + 4\varepsilon^2}), \qquad \varepsilon = 10^{-20} \tag{12.13}$$

Applied first to ρ, then to the internal energy $e_{\text{int}} = \rho E - \frac{1}{2}\rho v^2$.

CFL condition (polar grid)

$$\Delta t = C_{\text{CFL}}\ \min_{i,j}\left[\min\left(\frac{\Delta r_i}{|v_r| + c_s},\ \frac{r_i\,\Delta\theta_j}{|v_\theta| + c_s}\right)\right] \tag{12.14}$$

Atmosphere cells ($\rho_0 < 10^{-6}\rho_{\text{max}}$) are excluded from the CFL computation.

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§13 Strang Cartesian Solver

The Strang solver operates on a uniform Cartesian grid (x, y), $[0, L_x]\times[0, L_y]$, with cell sizes $\Delta x = L_x/N_x$, $\Delta y = L_y/N_y$ and $n_g = 2$ ghost layers.

Perturbation storage

The solver stores perturbations from a 1-D isentropic HSE background:

$$\rho'_{ij} = \rho_{ij} - \bar\rho(y_j), \qquad E'_{ij} = E_{ij} - \frac{\bar p(y_j)}{\gamma - 1} \tag{13.1}$$

Momentum $(\rho u, \rho v)$ is stored in total form (background velocity is zero).

Isentropic HSE background

$$\bar{\rho}(y) = \left[\rho_0^{\gamma-1} - \frac{(\gamma-1)\,g\,y}{\gamma\,K}\right]^{1/(\gamma-1)}, \qquad \bar{p}(y) = K\,\bar{\rho}(y)^{\gamma} \tag{13.2}$$

where ρ_0 is the base density, K the polytropic constant, g the uniform gravitational acceleration, and γ the adiabatic index.

MC (Monotonised Central) limiter

$$\mathrm{MC}(a,b) = \begin{cases} \mathrm{sgn}(a)\ \min\!\left(\frac{1}{2}|a+b|,\ 2|a|,\ 2|b|\right) & \text{if } ab > 0 \\ 0 & \text{otherwise} \end{cases} \tag{13.3}$$

Well-balanced y-sweep MUSCL reconstruction

Slopes are computed on perturbation quantities; face states are restored using face-centred HSE values:

$$\sigma_j^{\rho} = \mathrm{MC}(\rho'_j - \rho'_{j-1},\ \rho'_{j+1} - \rho'_j) \tag{13.4a}$$

$$\sigma_j^{P} = \mathrm{MC}(P'_j - P'_{j-1},\ P'_{j+1} - P'_j), \qquad P'_j = P_j - \bar{p}(y_j) \tag{13.4b}$$

$$\rho^{L}_{j+1/2} = \bar{\rho}(y_{j+1/2}) + \rho'_j + \tfrac{1}{2}\sigma_j^{\rho} \tag{13.4c}$$

$$P^{L}_{j+1/2} = \bar{p}(y_{j+1/2}) + P'_j + \tfrac{1}{2}\sigma_j^{P} \tag{13.4d}$$

The x-sweep uses no WB (background is y-dependent only); slopes and reconstruction are on total variables.

MUSCL-Hancock predictor

Both left and right face states of a cell are evolved by half a time step using the within-cell flux difference:

$$\mathbf{U}_{\text{face}}^{n+1/2} = \mathbf{U}_{\text{face}}^{n} + \frac{\Delta t}{2\Delta x}\left[\mathbf{F}(\mathbf{W}_L) - \mathbf{F}(\mathbf{W}_R)\right] \tag{13.5}$$

where $\mathbf{W}_L, \mathbf{W}_R$ are the MUSCL-reconstructed left and right primitives of the same cell. In the y-sweep, the gravity source $-\rho g$ is included in the Hancock half-step to balance $\partial\bar{p}/\partial y$:

$$(\rho v)_{\text{face}}^{n+1/2} \mathrel{+}= -\tfrac{1}{2}\Delta t\,\rho\,g \tag{13.6}$$

Strang splitting

$$\mathbf{U}^{n+1} = \mathcal{L}_x\!\left(\tfrac{\Delta t}{2}\right) \circ \mathcal{L}_y\!\left(\tfrac{\Delta t}{2}\right) \circ \mathcal{L}_y\!\left(\tfrac{\Delta t}{2}\right) \circ \mathcal{L}_x\!\left(\tfrac{\Delta t}{2}\right)\,\mathbf{U}^{n} \tag{13.7}$$

Each $\mathcal{L}$ is a MUSCL-Hancock + LM-HLLC unsplit sweep in the corresponding direction.

Boundary conditions (Cartesian)

x-periodic (n_g ghost layers):

$$U_{-g,\,j} = U_{N_x-g,\,j}, \qquad U_{N_x+g,\,j} = U_{g,\,j} \tag{13.8a}$$

y-bottom reflective ($y = 0$):

$$\rho_{i,-g} = \rho_{i,g-1}, \quad (\rho u)_{i,-g} = (\rho u)_{i,g-1}, \quad (\rho v)_{i,-g} = -(\rho v)_{i,g-1}, \quad E_{i,-g} = E_{i,g-1} \tag{13.8b}$$

y-top outflow (zero-gradient):

$$U_{i,\,N_y+g} = U_{i,\,N_y-1} \tag{13.8c}$$

CFL condition (Cartesian)

$$\Delta t = C_{\text{CFL}} \ \min_{i,j} \left[\min\left(\frac{\Delta x}{|u| + c_s},\ \frac{\Delta y}{|v| + c_s} \right) \right] \tag{13.9}$$

---

§14 Low-Mach HLLC (LM-HLLC)

The LM-HLLC Riemann solver modifies the standard HLLC (§4) to reduce excessive numerical diffusion at low Mach number. The modification scales the pressure jump in the contact wave speed estimate.

Local Mach number

$$M_{\text{local}} = \frac{|u_L| + |u_R|}{c_L + c_R} \tag{14.1}$$

Blending factor

$$f(M) = \text{clamp}(M_{\text{local}},\ M_{\text{cutoff}},\ 1) \tag{14.2}$$

with $M_{\text{cutoff}} = 10^{-3}$. At $M \gg M_{\text{cutoff}}$, $f = 1$ and standard HLLC is recovered. At $M < M_{\text{cutoff}}$, $f = M_{\text{cutoff}}$ and the acoustic dissipation is suppressed.

Modified contact wave speed

In the Strang solver:

$$S^* = \frac{f(M)(P_R - P_L) + \rho_L u_L(S_L - u_L) - \rho_R u_R(S_R - u_R)}{\rho_L(S_L - u_L) - \rho_R(S_R - u_R)} \tag{14.3}$$

In the FAS solver (equivalent, alternative formulation):

$$P_{L,\text{mod}} = \bar{P} + \tfrac{1}{2} f(P_L - P_R), \qquad P_{R,\text{mod}} = \bar{P} - \tfrac{1}{2} f(P_L - P_R) \tag{14.4a}$$

$$S^* = \frac{P_{R,\text{mod}} - P_{L,\text{mod}} + \rho_L u_L(S_L - u_L) - \rho_R u_R(S_R - u_R)}{\rho_L(S_L - u_L) - \rho_R(S_R - u_R)} \tag{14.4b}$$

where $\bar{P} = \frac{1}{2}(P_L + P_R)$. The two formulations are algebraically identical.

Physical interpretation

At low Mach number, the pressure jump across a contact wave is $O(M^2 P)$ while the acoustic pressure fluctuation is $O(MP)$. Standard HLLC treats both at the same order, creating $O(1/M)$ numerical dissipation. Scaling by $f(M)$ reduces the effective pressure jump, suppressing the spurious acoustic mode and allowing buoyancy-driven flows to develop correctly.

Consequence: sound waves with $M < M_{\text{cutoff}}$ are damped. For stellar convection ($M \sim 10^{-3}$–10^{-1}), this is acceptable because the dynamics are buoyancy-dominated.

---

§15 Cartesian Compatible Lagrangian (`cart_lag`)

Staggered quadrilateral mesh on $[0, L_x] \times [0, L_y]$. Cell-centered thermodynamic variables $(\rho, P, Q, e_{\text{int}})$; node-centered kinematic variables $(X, Y, v_x, v_y, F_x, F_y)$. Subcell forces distribute the cell's pressure + AV work to its 4 corner nodes, then `atomicAdd` to the node force buffer. Energy is updated from node displacements dotted with subcell forces (Caramana-Shashkov-Whalen 1998, JCP 144) — **compatible** in the sense that cell internal energy change equals the work that the cell does on its 4 corner nodes, to the bit.

Node mass

$$m_n = \tfrac{1}{4} \sum_{c \text{ adj } n} m_c \tag{15.1}$$

Under periodic BC the adjacent-cell wrap uses modular indexing; node copies at `in = nnx-1` and `jn = nny-1` receive `m_n` equal to their origin (0-indexed) partner, because they each independently iterate over the same wrap set. **Diagnostics must skip those duplicates** when computing $\sum m_n$ to avoid double counting (see P31).

Cell area via shoelace (for pre-Lagrangian volume)

$$V_c = \tfrac{1}{2}\left|\sum_{k=0}^{3} X_k Y_{k+1} - X_{k+1} Y_k\right| \tag{15.2}$$

indices modulo 4.

Divergence-consistent strain rate

$$s = -\frac{1}{V_c}\sum_{\text{edges}} \tfrac{1}{2}(v_a + v_b)\cdot \hat{n}_{ab}\,\ell_{ab} \tag{15.3}$$

Compression-only (`s > 0`) triggers artificial viscosity; dilation `s < 0` yields $Q = 0$.

von Neumann–Richtmyer artificial viscosity

$$Q = \rho\left(C_{\text{quad}}\,(sL)^2 + C_{\text{lin}}\,c_s\,sL\right)\cdot w_{\text{shear}} \tag{15.4}$$

with $L = \sqrt{V_c}$. The shear weight $w_{\text{shear}} \in [0, 1]$ (default 1) optionally reduces Q in shear-dominated cells using an estimate of $\partial u/\partial x$, $\partial v/\partial y$ from corner velocity differences (`--shear-aware-av`). Defaults $C_{\text{quad}} = 2$, $C_{\text{lin}} = 0.5$.

Subcell edge normal force (per cell, per edge k)

$$\mathbf{a}_k = (P + Q)\,(\Delta y_k,\ -\Delta x_k), \qquad \Delta x_k = X_{k+1} - X_k,\ \Delta y_k = Y_{k+1} - Y_k \tag{15.5}$$

Corner subcell force (contributes to node via `atomicAdd`)

$$\mathbf{F}^{(k)}_{\text{sub}} = \tfrac{1}{2}(\mathbf{a}_{k-1} + \mathbf{a}_k) \tag{15.6}$$

indices modulo 4. For uniform P on a Cartesian cell the four corner subcell forces sum to zero around every interior node — this is the basis of the uniform-advection invariance check (P30, P31).

Kick-drift-kick node update

$$\begin{aligned} \tfrac{1}{2}\Delta v &= \frac{F_n}{m_n}\tfrac{1}{2}\Delta t \\ \Delta X &= (v_n + \tfrac{1}{2}\Delta v)\Delta t \\ v_n^{n+1} &= v_n^n + \Delta v \end{aligned} \tag{15.7}$$

Compatible energy update (per cell)

$$m_c \Delta e_c = -\sum_{k=0}^{3} \mathbf{F}_{\text{sub}}^{(k)} \cdot \Delta X_k^{(\text{node})} \tag{15.8}$$

Energy removed from node KE equals energy deposited in cell IE, modulo machine epsilon.

Minimum-height CFL

$$\Delta t = \text{CFL} \cdot \min_c \frac{h_c}{c_s + |\mathbf{v}|} \cdot \min\left(1, \frac{1}{f_{\text{comp}}\, sL/c_s}\right) \tag{15.9}$$

where h_c is the minimum perpendicular height in the (possibly deformed) quadrilateral, and f_{comp} is a compression-strain safety fraction (default 0.25) to avoid grid inversion under strong shocks.

---

§16 Eulerian Rezone + Swept-Edge Remap (`cart_ale`, `cart_ale2`)

ALE step = Lagrangian substep (§15) → rezone → remap. Rezone: snap every node back to its uniform Cartesian position (X_0, Y_0). Remap: transfer cell-centered conserved quantities $\{dm, dm \cdot e_{\text{int}}, p_x, p_y\}$ across each edge via the **signed swept area** between the Lagrangian- displaced edge and the original edge.

Swept-region signed area (east face of cell (i, j))

$$A_{\text{sw}} = \tfrac{1}{2} \sum_{k=0}^{3} (X_k Y_{k+1} - X_{k+1} Y_k), \quad \text{vertices } (A_{\text{old}}, A_{\text{new}}, B_{\text{new}}, B_{\text{old}}) \tag{16.1}$$

$A_{\text{sw}} > 0$: sweep moved into cell R (donor = L); $A_{\text{sw}} < 0$: sweep moved into cell L (donor = R).

First-order donor-cell flux

$$\Delta f = \frac{f_{\text{donor}}}{V_{\text{donor}}} |A_{\text{sw}}|, \quad f \in \{dm,\ dm \cdot e_{\text{int}},\ p_x,\ p_y\} \tag{16.2}$$

with a safety clamp $|A_{\text{sw}}|/V_{\text{donor}} \leq 1/2$.

MUSCL 2nd-order reconstruction (Kucharik–Shashkov 2012)

Per-cell limited slopes on the reference mesh:

$$\partial_x f_c = \phi\left(\frac{f_{c+\hat{x}} - f_c}{\Delta x},\ \frac{f_c - f_{c-\hat{x}}}{\Delta x}\right) \tag{16.3}$$

same for $\partial_y f_c$. Limiter ϕ: `minmod`, van Leer (default), or MC (`--remap-limiter`). Face-centroid evaluation:

$$\hat{f}(\xi, \eta) = f_c + \partial_x f_c \cdot (\xi - x_c) + \partial_y f_c \cdot (\eta - y_c) \tag{16.4}$$

evaluated at the swept-region centroid $(\xi, \eta) = \frac{1}{4} \sum A_k$.

Remap update

$$\Delta f_{\text{donor}} = -\Delta f, \quad \Delta f_{\text{acceptor}} = +\Delta f \tag{16.5}$$

Discrete conservation is **exact** because the same flux value is subtracted from donor and added to acceptor.

Periodic BC wrap edge

Under `--bc-x periodic`, the east-face edge $ic = n_x - 1$ is **not** a wall but a wrap connecting cell $n_x - 1$ with cell 0. Remap kernels extend their edge count from $(n_x - 1) \cdot n_y$ to $n_x \cdot n_y$ (and similarly in y), with `cR_idx = (ic + 1) \bmod n_x`. The face-centroid-relative coordinate $s_x = (c_x - x_{\text{donor}})/\Delta x$ must be wrapped: if $s_x > 0.5$ then $s_x \mathrel{-}= n_x$, else if $s_x < -0.5$ then $s_x \mathrel{+}= n_x$ (see P30).

Node velocity rebuild (mass-weighted)

Post-remap node velocity is built from new cell-centered momentum and mass density:

$$v_n = \frac{\sum_{c \text{ adj } n} \frac{1}{4} p_c}{\sum_{c \text{ adj } n} \frac{1}{4} m_c} \tag{16.6}$$

Momentum-conservative: $\sum_n m_n v_n = \sum_c p_c$ to machine precision.

---

§17 Cart_ale2 PPM Reconstruction Variants

Extends §16 to piecewise-parabolic remap (Colella-Woodward 1984) with optional extremum-preserving limiter (Colella-Sekora 2008), primitive- variable space, and characteristic-variable projection (Stone et al. 2008 Appendix A, adiabatic hydro branch of `athena/reconstruct/ppm.cpp`).

PPM 4-point interpolant

$$f_{i+1/2} = \tfrac{1}{12}\big(7(f_i + f_{i+1}) - (f_{i-1} + f_{i+2})\big) \tag{17.1}$$

Yielding initial left/right face values $f_L = f_{i-1/2}$, $f_R = f_{i+1/2}$.

Classical CW monotonization

$$\begin{aligned}&\text{if } (f_L - f_c)(f_c - f_R) \le 0: \; f_L = f_R = f_c\\&\text{else if } \Delta f \cdot \Delta_6 > (\Delta f)^2: \; f_L = 3f_c - 2f_R\\&\text{else if } \Delta f \cdot \Delta_6 < -(\Delta f)^2: \; f_R = 3f_c - 2f_L\end{aligned} \tag{17.2}$$

with $\Delta f = f_R - f_L$, $\Delta_6 = 6(f_c - \frac{1}{2}(f_L + f_R))$. CW clamps smooth extrema (penalizes KH roll-up).

Colella-Sekora extremum-preserving limiter

Two stages (`--ppm-limiter cs`, default):

Stage 1 — face-value correction at smooth extrema, using five-point second-derivative stencils: $d^2_{i-1} = f_{i-2} + f_i - 2f_{i-1}$ and likewise d^2_i, d^2_{i+1}. At face f_L (i.e. $f_{i-1/2}$), if $(f_L - f_{i-1})(f_i - f_L) < 0$:

$$f_L \leftarrow \tfrac{1}{2}(f_{i-1} + f_i) - q_d/6 \tag{17.3}$$

where $q_a = 3(f_{i-1} + f_i - 2f_L)$ and

$$q_d = \begin{cases}\operatorname{sgn}(q_a)\ \min(C_2|d^2_{i-1}|, C_2|d^2_i|, |q_a|) & \text{if signs agree}\\ 0 & \text{otherwise}\end{cases} \tag{17.4}$$

with $C_2 = 1.25$. Analogous at f_R.

Stage 2 — parabolic coefficient limiting with smooth-vs-shock discrimination. Let $d^2_f = 6(f_L + f_R - 2f_c)$. If signs of $d^2_{i-1}, d^2_i, d^2_{i+1}, d^2_f$ all agree,

$$q_e = \operatorname{sgn}(d^2_f)\ \min(C_2|d^2_{i-1}|, C_2|d^2_i|, C_2|d^2_{i+1}|, |d^2_f|), \quad \rho_r = q_e/d^2_f \tag{17.5}$$

At a local extremum $(f_c - f_L)(f_R - f_c) \le 0$ and $\rho_r < 1 - \epsilon$: scale parabola smoothly ($f_L \leftarrow f_c - \rho_r(f_c - f_L)$, $f_R \leftarrow f_c + \rho_r(f_R - f_c)$). Otherwise apply classical CW overshoot clamp. The result: smooth peaks survive, true shocks still clamp.

Primitive vs conservative variable space

`--ppm-space cons`: reconstruct the 4 conserved densities $\{\rho, \rho e_{\text{int}}, \rho v_x, \rho v_y\}$. Unstable on smooth shear layers where ρv_x crosses zero (overshoot → negative mass × negative velocity → negative pressure).

`--ppm-space prim` (default): reconstruct the 4 primitives $\{\rho, P, v_x, v_y\}$. At swept centroid evaluate primitives then convert to conserved flux:

$$\begin{aligned}\Delta\widehat{dm} &= \hat{\rho}\,|A_{\text{sw}}|\\ \Delta\widehat{p_x} &= \hat{\rho}\,\hat{v}_x\,|A_{\text{sw}}|\\ \Delta\widehat{p_y} &= \hat{\rho}\,\hat{v}_y\,|A_{\text{sw}}|\\ \Delta\widehat{ie} &= \hat{P}\,|A_{\text{sw}}|/(\gamma - 1)\end{aligned} \tag{17.6}$$

The same flux value is subtracted from donor and added to acceptor, preserving discrete conservation exactly.

Characteristic projection (default on, `--ppm-char`)

Before PPM reconstruction, project the 5-point primitive stencil into local characteristic variables using cell-i's (ρ_i, a_i) where $a_i = \sqrt{\gamma P_i/\rho_i}$. For x-direction sweep:

$$\begin{aligned} w_0 &= \tfrac{1}{2}(P/a^2 - \rho_i\, v_x/a_i) && \text{(left acoustic)} \\ w_1 &= \rho - P/a^2 && \text{(entropy)} \\ w_2 &= v_y && \text{(shear)} \\ w_3 &= \tfrac{1}{2}(P/a^2 + \rho_i\, v_x/a_i) && \text{(right acoustic)} \end{aligned} \tag{17.7}$$

Apply PPM + CS limiter to each w_n independently. Project back using the same (ρ_i, a_i) basis:

$$\begin{aligned} \rho_f &= w_0^f + w_1^f + w_3^f \\ v_x^f &= a_i(w_3^f - w_0^f)/\rho_i \\ v_y^f &= w_2^f \\ P_f &= a_i^2(w_0^f + w_3^f) \end{aligned} \tag{17.8}$$

For y-direction sweep, swap the role of v_x and v_y in the projection (i.e. pass v_y where v_x appears in Eq. 17.7 and vice versa), then unpack with the same swap.

Rationale: in a smooth shear layer, acoustic and shear modes are independent — projecting and limiting each mode separately prevents shear gradients from triggering pressure overshoots. Mirrors Athena `ppm.cpp:99-107` + `characteristic.cpp:235-256, 470-492`.

KH benchmark (reality check)

The stack (17.1)–(17.8) keeps the Lecoanet canonical KH (Athena iprob=4 geometry, $k = 1$, $v_{\text{flow}} = 1$, $P_0 = 10$) stable to $t = 5$ at 256×512, with IE conserved to 10 digits and total E drift ~1.2%. **However**: the kinetic-energy spectrum at $k = 7$, 512×1024 falls as $\sim k^{-10}$ past the injection scale, not k^{-3}. The effective Reynolds number of this ALE stack is bounded by the Caramana subcell force dissipation, which is tens of times stronger than an HLLC face flux. `cart_ale2` is therefore a **compressible stellar convection / PdV pulsation** tool, not a 2D turbulence benchmark tool. For Kraichnan cascade studies use the `pseudo_spectral` branch.

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§18 Pseudo-Spectral Incompressible NS (`pseudo_spectral`)

2D doubly-periodic Navier–Stokes in vorticity–streamfunction form, solved on a uniform (N_x, N_y) grid by cuFFT R2C/C2R with Orszag skew-symmetric convection + IFRK3 integrating-factor viscosity + circular 2/3 dealiasing. See `docs/pseudo_spectral_design_2026-05-01.md` for benchmarking and background.

Vorticity–streamfunction primitives

$$\frac{\partial \omega}{\partial t} + \mathbf{u} \cdot \nabla \omega = \nu \, \nabla^2 \omega, \qquad \nabla^2 \psi = -\omega, \qquad u = \partial_y \psi, \;\; v = -\partial_x \psi. \tag{18.1}$$

In spectral space the Poisson solve is diagonal: $\hat{\psi}(\mathbf{k}) = \hat{\omega}(\mathbf{k})/|\mathbf{k}|^2$ (with the $\mathbf{k} = 0$ mode set to zero to fix the constant gauge).

Circular 2/3 dealiasing

Let $k_{\text{max}} = \min(N_x, N_y)\pi/L$ (assuming $L_x = L_y = L$). The dealias mask zeroes out all modes with $|\mathbf{k}| > \frac{2}{3} k_{\text{max}}$.

$$\hat{\omega}(\mathbf{k}) \leftarrow \begin{cases} \hat{\omega}(\mathbf{k}) & |\mathbf{k}| \le \frac{2}{3} k_{\text{max}} \\ 0 & \text{otherwise} \end{cases} \tag{18.2}$$

A circular rather than axis-aligned truncation avoids anisotropic under-resolution in the corners of **k**-space.

Orszag skew-symmetric convection

Two mathematically equivalent forms of $\mathbf{u} \cdot \nabla \omega$:

$$N_A(\omega) \equiv \mathbf{u} \cdot \nabla \omega, \qquad N_C(\omega) \equiv \nabla \cdot (\mathbf{u} \, \omega) \tag{18.3}$$

agree for $\nabla \cdot \mathbf{u} = 0$ in the continuum. After FFT truncation the two discretisations differ by $\mathcal{O}(\Delta k)$ terms that act as numerical dissipation. Orszag (1971) shows the symmetrised form $N_S = \frac{1}{2}(N_A + N_C)$ is an anti-symmetric operator in the truncated spectral representation, so it exactly conserves both enstrophy $\sum |\hat{\omega}|^2$ and kinetic energy $\sum |\hat{\omega}|^2/|\mathbf{k}|^2$ at the discrete level.

$$\hat{N}_S(\mathbf{k}) = \tfrac{1}{2} \big[\hat{N}_A(\mathbf{k}) + \hat{N}_C(\mathbf{k}) \big] \tag{18.4}$$

CLI flag `--ps-adv-only` substitutes the advective-only form $\hat{N}_A$ for reference.

Integrating-factor RK3 (IFRK3)

Substitute $\hat{\omega}(\mathbf{k}, t) = e^{-\nu |\mathbf{k}|^2 t} \, \hat{w}(\mathbf{k}, t)$. The viscous term is absorbed exactly; the evolution equation for $\hat{w}$ reads

$$\frac{\hat{w}}{t} = e^{+\nu |\mathbf{k}|^2 t} \left[-\hat{N}_S(\hat{\omega}) \right] \tag{18.5}$$

Integrated by Shu–Osher SSP-RK3 with integrating-factor weights $E \equiv e^{-\nu |\mathbf{k}|^2 \Delta t}$:

$$
\begin{aligned}
\widehat{\omega}_1 &= E\,\widehat{\omega}_n + \Delta t\,E\,[-\widehat{N}_S(\widehat{\omega}_n)],\\
\widehat{\omega}_2 &= \tfrac{3}{4}E^{1/2}\widehat{\omega}_n + \tfrac{1}{4}E^{-1/2}\widehat{\omega}_1 + \tfrac{1}{4}\Delta t\,E^{-1/2}\,[-\widehat{N}_S(\widehat{\omega}_1)],\\
\widehat{\omega}_{n+1} &= \tfrac{1}{3}E\,\widehat{\omega}_n + \tfrac{2}{3}E^{1/2}\widehat{\omega}_2 + \tfrac{2}{3}\Delta t\,E^{1/2}\,[-\widehat{N}_S(\widehat{\omega}_2)].
\end{aligned}
\tag{18.6}
$$

Viscous stability becomes unconditional (viscosity is mode-by-mode exact), so Δt is set solely by the advective CFL $\Delta t \le C_{\text{CFL}}\,\Delta x/\max|\mathbf{u}|$, $C_{\text{CFL}} \approx 0.5$.

Effective viscosity diagnostic

The actual dissipation rate of enstrophy $\mathcal{E} \equiv \frac{1}{2}\sum|\widehat{\omega}|^2$ is compared against the prescribed ν via

$$
\nu_{\text{eff}} \equiv -\frac{\mathcal{E}/t}{2\sum|\mathbf{k}|^2\,|\widehat{\omega}(\mathbf{k})|^2}. \tag{18.7}
$$

A healthy run has $\nu_{\text{eff}}/\nu \to 1$ in the statistically steady regime. Deviation signals numerical dissipation leakage (typically from insufficient dealiasing or Δt too large).

Energy spectrum (ring average)

$$
E(k) = \frac{1}{2}\sum_{k-\frac{1}{2}<|\mathbf{k}'|\le k+\frac{1}{2}} \frac{|\widehat{\omega}(\mathbf{k}')|^2}{|\mathbf{k}'|^2} \tag{18.8}
$$

with k binned in integer shells. The forced-turbulence benchmark (`--test forced_turb`) shows a clean k^{-3} enstrophy cascade past the injection scale and a modest $k^{-5/3}$ inverse-cascade tail.

---

§19 Sturm–Liouville Spectral Basis (Phase 0 ext+ — 1D offline)

This section documents the 1D Python solvers developed during Phase 0 ext+ (2026-05-02..03) to certify the spectral approach for the reduced-pressure Poisson problem and the GYRE adiabatic pulsation eigenvalue problem. These routines do **not** ship in the GPU solver — they are regression oracles and building blocks for the Phase 1 Fourier–Chebyshev 2D solver. The authoritative derivation of the spectral formulation and its convergence analysis is `docs/spectral_stratified_poisson_report_2026-05-03.md`; this section records the code-level contract.

Reduced-pressure Poisson operator (Fourier in x)

Writing the reduced pressure $\pi = p/\rho_0$, the variable-density Poisson equation reads $\nabla\cdot(\rho_0\nabla\pi) = \tilde{f}$, and its Fourier mode $\widehat{\pi}(k_x, r)$ satisfies

$$
\frac{}{r}\left[\rho_0(r)\,\frac{\widehat{\pi}}{r}\right] - k_x^2\,\rho_0(r)\,\widehat{\pi} = \widehat{\tilde{f}}(k_x, r), \qquad r \in [a, b]. \tag{19.1}
$$

Chebyshev–Gauss–Lobatto collocation

On the CGL grid $\xi_j = \cos(j\pi/N)$, $j = 0, \dots, N$, mapped affinely to $r \in [a, b]$, and with D the Trefethen spectral differentiation matrix (size $(N+1) \times (N+1)$):

$$L_N = D\, R_\rho\, D - k_x^2\, R_\rho, \qquad R_\rho \equiv \mathrm{diag}(\rho_0(r_j)). \tag{19.2}$$

Dirichlet BCs are imposed by strong collocation (unit rows at $j = 0, N$).

Convergence regime (integer vs fractional surface exponent)

For $\rho_0(r) \sim (R - r)^\sigma$ near the surface:

$$\|\pi_N - \pi_{\text{exact}}\|_\infty \sim \begin{cases} \rho^N & \sigma \in \mathbb{Z}_{\geq 0} \quad \text{(spectral)} \\ N^{-\sigma-1/2} & \sigma \notin \mathbb{Z} \quad \text{(algebraic, Trefethen Thm 7.2)} \end{cases} \tag{19.3}$$

Lane–Emden $n = 3$ (Eddington, $\sigma = 3$) is the only standard physically-motivated polytrope in the spectral-convergence branch; $n = 3/2$ (convective core) is in the N^{-2} algebraic branch and requires a Jacobi-weighted basis (outside the current implementation).

Liouville potential (reference-only, not used in Phase 1)

Under $\hat{\pi} = \rho_0^{-1/2}\, q$, equation (19.1) transforms to

$$q'' + \widetilde{W}(r)\, q - k_x^2\, q = \tilde{g}, \qquad \widetilde{W} = \frac{\rho_0''}{2\rho_0} - \frac{(\rho_0')^2}{4\rho_0^2}, \tag{19.4}$$

exhibiting the k_x-independent Schr”odinger structure. Near the surface $\widetilde{W} \sim \sigma(\sigma-2)/[4(R-r)^2]$. The Liouville programme’s original promise — “one basis diagonalises Poisson and the g-mode operator simultaneously” — fails in the strong sense: the Poisson operator’s optimal prefactor is $\alpha_\star(\text{Poisson}) = 1 - \sigma/2$ whereas the g-mode operator has $\beta_\star(\text{g-mode}) = \ell + 1$ (different singular point, $r = 0$ vs $r = R$). The operational consequence is that the Chebyshev mesh is shared but the operators are assembled and factorised independently; see the technical report §7 for the full analysis.

GYRE 4-variable adiabatic pulsation equations

For ℓ-mode oscillations of a non-rotating star, with $\alpha_{\text{grv}} = 1$ (full self-gravity), using the Unno et al. (1989) non-dimensional variables (y_1, y_2, y_3, y_4):

$$\begin{aligned} x\,\tfrac{y_1}{x} &= (V_g - \ell - 1)\, y_1 + \left(\tfrac{\lambda}{c_1\omega^2} - V_g\right) y_2 + \tfrac{\lambda}{c_1\omega^2}\, y_3, \\ x\,\tfrac{y_2}{x} &= (c_1\omega^2 - A^\star_{\text{iso}})\, y_1 + (A^\star - U + 3 - \ell)\, y_2 - y_4, \\ x\,\tfrac{y_3}{x} &= (3 - U - \ell)\, y_3 + y_4, \\ x\,\tfrac{y_4}{x} &= U A^\star\, y_1 + U V_g\, y_2 + \lambda\, y_3 + (2 - U - \ell)\, y_4, \end{aligned} \tag{19.5}$$

with $x = r/R$, $\lambda = \ell(\ell+1)$, $y_3 = \Phi'/(gr)$, $y_4 = (\Phi'/r)/g$. Structure coefficients $V_g = V/\Gamma_1$, $A^\star$, U, c_1, Γ_1 follow the Unno et al. definitions and are read from the GYRE native polytropic file (`/tmp/gyre_run/poly3.txt` for Lane–Emden $n = 3$).

Regularity and vacuum boundary conditions

Regularity at $x = 0$ and vacuum at $x = 1$:

$$\begin{aligned} &\text{inner:} \quad c_1\omega^2 y_1 - \ell y_2 - \ell y_3 = 0, \qquad \ell y_3 - y_4 = 0, \\ &\text{outer:} \quad y_1 - y_2 = 0, \qquad U y_1 + (\ell+1) y_3 + y_4 = 0. \end{aligned} \tag{19.6}$$

Generalised eigenvalue problem and benchmark

After Chebyshev collocation on $x \in [0.01, 0.999]$ with $N+1$ nodes, (19.5) assembles into a $4(N+1) \times 4(N+1)$ generalised eigenproblem

$$\mathsf{P}\,\mathbf{u} = \omega^{-2}\,\mathsf{Q}\,\mathbf{u}, \tag{19.7}$$

in which the inverse-ω^2 structure of equation (19.5a) requires the generalised (linearised) form rather than a standard linear eigenproblem. At $N = 48$ (192 DOF) the first radial-order g-mode frequency $\omega^2_{n_g=1}(\ell = 1)$ agrees with the GYRE reference to 5.9×10^{-7}; the maximum relative error over the first ten radial orders is 1.5×10^{-6}. A $21\times$ reduction in DOF and $350\times$ lower maximum error compared with a staggered second-order finite-difference discretisation at $N_r = 1024$.

Barycentric Lagrange evaluation

The $N + 1$ Chebyshev coefficients $\{a_n\}$ or nodal values $\{u_j\}$ define a continuous representation that is evaluable at any $r^\star$ via the barycentric formula

$$u(r^\star) = \frac{\sum_j w_j\,u_j/(r^\star - r_j)}{\sum_j w_j/(r^\star - r_j)}, \qquad w_j = (-1)^j\,c_j, \tag{19.8}$$

with $c_0 = c_N = 1/2$, $c_j = 1$ otherwise (Berrut & Trefethen 2004). The evaluation is rounding-error stable, with error $\leq 10^{-12}$, and is the mechanism by which a low-N Chebyshev representation supports arbitrary-resolution output rendering.

Reproducibility

Frozen regression oracles (`python <script>.py --verify` exits zero):

- `scripts/gmode_exp_i_gyre_compat.py` — 2-var Cowling FD, GYRE Cowling benchmark, max rel. diff. 5.6×10^{-4} at $N_r = 1024$.
- `scripts/gmode_exp_j_full_gyre_compat.py` — 4-var full gravity FD, GYRE full benchmark, max rel. diff. 5.3×10^{-4}.

- `scripts/gmode_exp_k_chebyshev_full.py` — 4-var full gravity Chebyshev, GYRE full benchmark, max rel. diff. 1.5×10^{-6} at $N = 48$.
- `scripts/spectral_analytical_ceiling.py` — three analytic ceiling tests (manufactured Poisson, harmonic oscillator, Dirichlet Laplacian); all reach 10^{-13} to 10^{-15} at $N \lesssim 64$.

The continuous-representation demonstration (`scripts/spectral_resolution_demo.py`) shows the $N = 48$ Chebyshev basis produces the same eigenfunction at 4096-point barycentric sampling as the FD $N_r = 1024$ solution at the same points, to within the $N = 48$ discretisation error of 3.4×10^{-3}.

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1.3	Sound speed	`eos.h`, `strang_device.cuh`, `fas_hllc.cuh`
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1.5	Radial flux	`hydro/riemann.cpp`, `fas_hllc.cuh`
1.6	Theta flux	`hydro/riemann.cpp`, `fas_hllc.cuh`
1.7	Geometric + gravity source	`hydro/flux.cpp`, `fas_residual.cu`
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Eq.	Description	Source file(s)
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